UNIT
2

FLOWERING AND NON-FLOWERING PLANTS

Learning objectives:

- Differentiate between flowering and non-flowering plants.
- Identify and describe the characteristics of flowering and non-flowering plants.
- Understand the life cycle of flowering plants, from seed to seed dispersal.
- Recognize the importance of flowering and non-flowering plants in the environment.

As living organisms are divided into different classes, plants are grouped into two main types: **flowering plants** and **non-flowering plants**. Let's learn about their characteristics and the life cycle of flowering plants.

Flowering Plants:

Flowering plants are those that produce flowers. They are also known as **Angiosperm** (means "covered seed"). These flowers play a key role in reproduction by forming seeds that grow into new plants. Some examples of flowering plants include roses, mango trees, and sunflowers.

Characteristics of Flowering Plants:

- They produce flowers that may develop into fruits.
- They reproduce through seeds.
- They have roots, stems, leaves, and flowers.
- Most of them are green and make their food through photosynthesis.
- They are found in a variety of environments, such as gardens, forests, and fields.
- Fruits and vegetables that we eat, like apples and tomatoes, are parts of flowering plants.

Non-Flowering Plants

Non-flowering plants do not produce flowers. They are also known as **Gymnosperm** (means "naked seed"). They reproduce by spores or other means instead of seeds. Some examples of non-flowering plants are mosses, ferns, and algae.

Characteristics of Non-Flowering Plants:

- They do not produce flowers or seeds.
- They reproduce through spores or cones.
- Many of them prefer damp or shady places to grow.
- Non-flowering plants like algae are found in water, while ferns and mosses grow on land.
- These plants play an essential role in maintaining ecosystems by producing oxygen and preventing soil erosion.

DID YOU KNOW?

Conifer (A type of plant that produces cones and has needle-like leaves) are endangered creatures due to climate change, logging and insect harms.

Life Cycle of Flowering Plants

Flowering plants have a fascinating life cycle:

Seed: Every plant starts its life as a seed. The seed contains a small plant (embryo) and stored food.

Germination: With the right conditions of water, soil, and sunlight, the seed grows into a seedling.

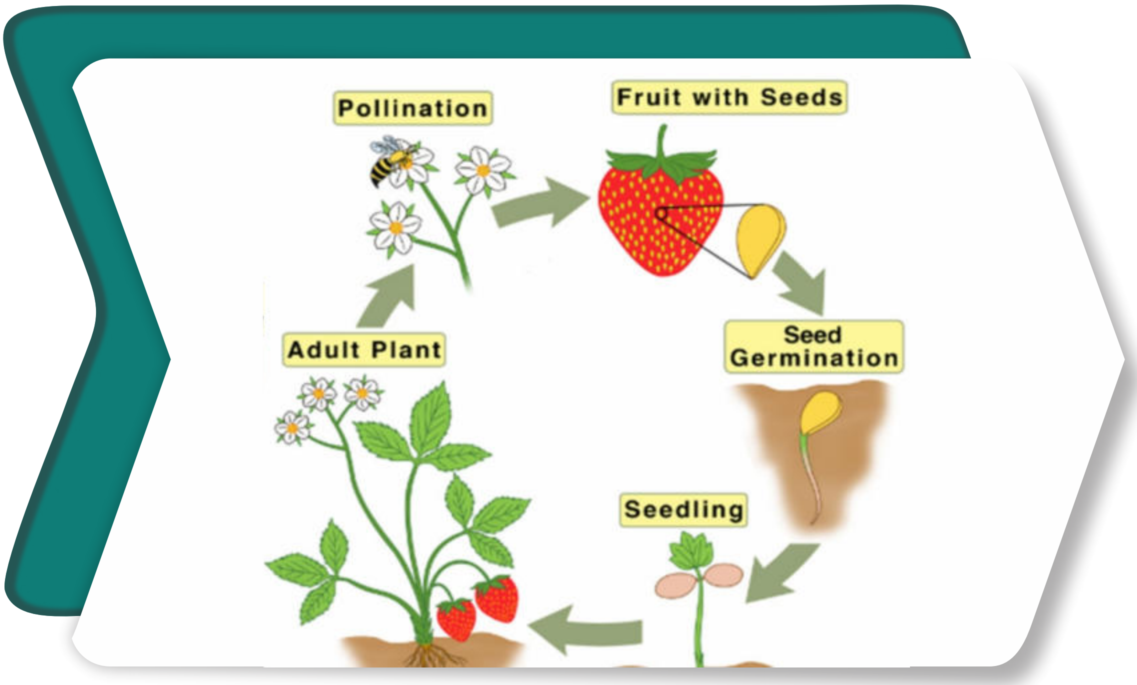
Growth: The seedling grows into a mature plant with leaves, roots, and stems.

Flowering: The mature plant produces flowers.

Pollination: Pollen (A fine powder that helps plants make seeds) from one flower reaches another flower with the help of wind,

Fertilization: Seeds develop inside the flower after pollination.

Seed Dispersal: Seeds are scattered by wind, water, or animals to grow into new plants.



Fun Fact

Did you know that mangoes, which are called the "king of fruits" in Pakistan, come from flowering plants? Similarly, wheat and rice, which are staple foods in Pakistan, are also flowering plants!

Exercises

Write down the name of plants into their correct category.


Moss


Daisy


Pine Tree


Fern


Tulip


Sunflower


Snake Plant


Rose


Conifer


Orchid

Flowering Plants	Non-flowering Plants

Fill in the blanks.

1. Plants that produce flowers are called ______________ plants.

2. Non-flowering plants reproduce through ______________ .

3. ______________ is the process by which a seed grows into a seedling.

4. Examples of flowering plants include ______________ and ______________ .

5. Non-flowering plants like ______________ grow in water.

6. Pollen is transferred from one flower to another during ______________ .

7. Seeds are dispersed by ______________ or ______________ .

8. The small plant inside a seed is called the ______________ .

Write "T" for True or "F" for False.

1. Non flowering plants are also called as angiosperm

2. Non-flowering plants do not produce seeds.

3. Wheat is an example of a non-flowering plant.

4. Pollination is necessary for the formation of seeds in flowering plants.

5. Algae and mosses grow mostly in dry areas.

6. Fruits are formed from flowers.

7. Seeds cannot grow without water.

8. Germination occurs before pollination.

Choose the correct answer.

1. Which part of a flowering plant produces seeds?

- Root
- Flower
- Stem
- Leaf

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2. Which of the following is a non-flowering plant?

- Sunflower
- Rose
- Fern
- Mango

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3. What is the first stage of a flowering plant's life cycle?

- Germination
- Seed
- Growth
- Flowering

4.Which process involves pollen transfer between flowers?

- Germination
- Pollination
- Fertilization
- Photosynthesis

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5. is an example of a flowering plant.

- Fern
- Algae
- Mango tree
- Moss

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6. Which part of the plant grows first during germination?

- Leaf
- Root
- Flower
- Stem

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7. Spores are used for reproduction by:

- Mango trees
- Ferns
- Sunflowers
- Roses

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8. What do flowers form after pollination and fertilization?

- Spores
- Fruits
- Roots
- Stems

Answer the following questions.

Q1. Write about the characteristics of flowering plants which non-flowering do not possess with example.

Q2. How do non-flowering plants reproduce?

Q3. What are spores? Name two non-flowering plants that reproduce through spores.

Q4. Why is pollination important for flowering plants?

Q5. Describe the life cycle of a flowering plant in your own words.

Q6. How are seeds dispersed in nature?

Q7. Why are plants important for life on Earth?

Critical Thinking

State your opinion: Can you save conifers from danger? What steps would you take to protect them? Do conifers store food in their roots?

Grow Your Own Plant:

Objective: To observe and compare the germination process of different seeds (flowering plants) under various conditions.

Materials:

- Different types of seeds (e.g., beans, sunflower, tomato seeds)
- Clear plastic cups or containers
- Paper towels
- Water
- Markers

Procedure:

- Place a damp paper towel in each cup.
- Place a few seeds of each type on the paper towels.
- Label each cup with the type of seed.
- Place one cup in a dark place, one in a sunny location, and one in a cool, dark place.
- Observe and record the changes daily for a week or two.

Questions:

- Which seeds germinated first?

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- How did the location affect the germination process?

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- What is the most interesting part of this experiment?

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